

Birla Institute of Technology and Science, Pilani

EEE F244 / INSTR F244 Microelectronic Circuits
Comprehensive Examination, II Semester 2015-2016

Part A: CLOSED BOOK

Time: 40 min.

Max. Marks=30

Date: 07-05-2016

Name:

ID:

Unless Given specifically

Take -- $V_{DD} = 3.3V$,

For NMOS device $\mu_n C_{ox} = 140 \mu A/V^2$, $V_T = 0.7 V$, $\lambda = 0.1 V^{-1}$, $C_{ox} = 0.38 pF/\mu m^2$

For PMOS device $\mu_p C_{ox} = 40 \mu A/V^2$, $V_T = -0.8 V$, $\lambda = 0.1 V^{-1}$, $C_{ox} = 0.38 pF/\mu m^2$

NOTE:

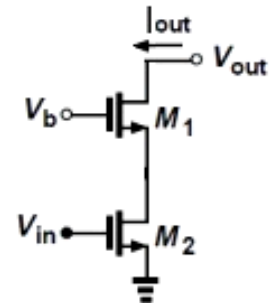
If not specified in question,-----

- Ignore γ , λ in drain current equation.
- Bulk of nmos connected to ground and bulk of pmos connected to V_{dd} . Specify your assumptions. Justify your answers.
- Unless specified, assume all MOSFET are biased in saturation region
- All symbols have usual meaning

***Write the answer to the point in the specified area (mentioned as 'Ans').**

***No partial marking.**

- (1) Two identical NMOS transistors M_1 and M_2 are connected as shown in figure. V_b is chosen such that both the transistors are in saturation. Find out the $\frac{dI_{out}}{dV_{in}}$ at constant V_{out} . [MM: 2]



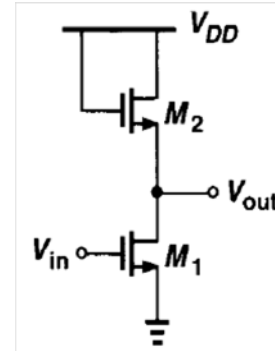
(Ans): $(dI_{out}/dV_{in}) = \text{-----}$

- (2) Find out the intrinsic gain ($g_m r_o$) of a NMOS transistor when W/L ratio is 100 and drain current is $0.5 \mu A$. [MM: 2]

(Ans): $g_m r_o = \text{-----}$

- (3) Find out the gain value (A_v) of the given circuit when $I_{D1}=I_{D2}$.

Consider, $\left(\frac{W}{L}\right)_{M1} = 100$, $\left(\frac{W}{L}\right)_{M2} = 10$ and $\eta=0.18$. [MM: 2]



(Ans): A_v -----

- (4) For a typical NMOS, assume $V_S=0.5$ V, $V_{TH0}=0.6$ V, $\gamma = 0.4$ V^{1/2} and $\phi = 0.4$ V. Find out the V_{TH} . Also find out the drain current value (I_D) when $V_G=V_D=1.4$ V, $\mu_n C_{OX}=100$ $\mu A/V^2$, $W/L=50$ and $\lambda=0$. [MM: 2+2=4]

(Ans): V_{TH} -----

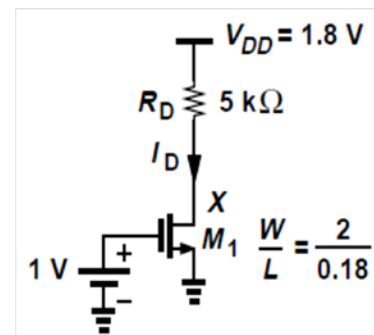
(Ans): I_D -----

- (5) A MOSFET carries a drain current of 1 mA with $V_{DS} = 0.5$ V in saturation. Determine the change in I_D (ΔI_D) if V_{DS} rises to 1 V. Consider $\lambda=0.1$ V⁻¹. Also find out the output impedance (r_0) of the device? [MM: 2+2=4]

(Ans): ΔI_D -----

(Ans): r_0 -----

- (6) Calculate the drain current (I_D) of M_1 transistor. Assume $\mu_n C_{OX} = 100$ $\mu A/V^2$ and $V_{TH} = 0.4$ V. Also determine the value of W/L when the M_1 transistor is at the edge of saturation. [MM: 2+2=4]

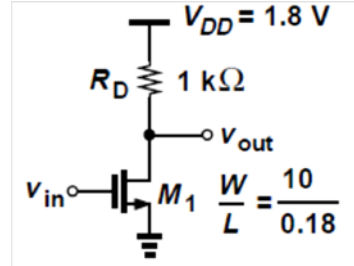


(Ans): $I_D = \dots\dots\dots$

(Ans): $(W/L)_{\max} = \dots\dots\dots$

- (7) Calculate the small signal voltage gain (A_v) value of the given amplifier circuit when M_1 transistor operates in saturation. Consider $V_{DS} = 0.8$ V, $V_{TH} = 0.5$ V, $\mu_n C_{OX} = 100 \mu A/V^2$ and $\lambda = 0$.

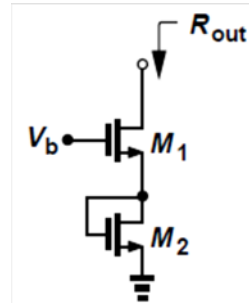
[MM: 3]



(Ans): $A_v = \dots\dots\dots$

- (8) Compute the output resistance of the given circuit with the help of small signal model. Consider that M_1 and M_2 are identical and operates in saturation.

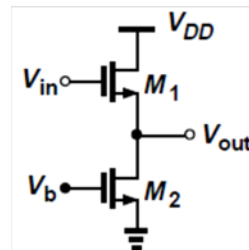
[MM: 3]



(Ans): $R_{out} = \dots\dots\dots$

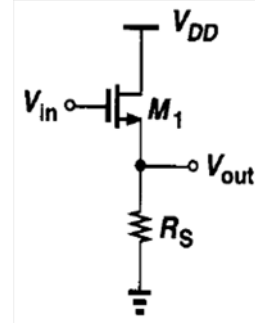
- (9) In the given circuit M_2 serves as a current source. Compute the small signal voltage gain of the circuit.

[MM: 3]



(Ans): $A_v = \dots\dots\dots$

- (10) In the given source follower circuit voltage gain is 0.5 to drive $50\ \Omega$ of source load (R_S). Find out the W/L ratio of M_1 when drain current through the circuit is $5.56\ \text{mA}$. Assume $\mu_n C_{OX} = 100\ \mu\text{A}/\text{V}^2$, $V_{TH} = 0.5\ \text{V}$, $\lambda = 0$ and $r_{O1} = \infty$ [MM: 3]



(Ans): $(W/L)_{M1} = \dots\dots\dots$

*****End*****

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Comprehensive Examination, II Semester 2015-2016

Part B: OPEN BOOK

Time: 140 min.

Max. Marks=80

Date: 07-05-2016

Unless Given specifically

Take -- $V_{DD} = 3.3V$,

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For PMOS device $\mu_p C_{ox} = 40 \mu A/V^2$, $V_T = -0.8 V$, $\lambda = 0.1 V^{-1}$, $C_{ox} = 0.38 pF/\mu m^2$

NOTE:

If not specified in question,-----

- Ignore γ , λ in drain current equation.
- Bulk of nmos connected to ground and bulk of pmos connected to V_{dd} . Specify your assumptions. Justify your answers.
- Unless specified, assume all MOSFET are biased in saturation region
- All symbols have usual meaning

Q.1 For the MOS amplifier shown in **Fig.1**, load is PMOS cascode load, all transistors have same $\mu_{n/p} C_{ox} W/L$ of $1400 \mu A/V^2$, $V_T = 0.7 V$, dc bias current of $100 \mu A$ and $\lambda = 0.1 V^{-1}$. Assume all transistors are in saturation.

- (a) Find the value of V_{b2} which gives a minimum output voltage. What is the value of minimum output voltage.
- (b) Compute the value of small-signal output resistance (R_{out}). Also find the small-signal low frequency voltage gain (V_{out}/V_{in}).
- (c) Find the expression of small-signal low frequency voltage gain (V_{out}/V_{in}) when from the circuit, transistor M_1 is removed and input voltage (V_{in}) is applied at the source of transistor M_2 .
- (d) Using the circuit of **Fig.1**, sketch and label the circuit of an NMOS cascode fully differential amplifier. Use current source I_{ss} for current sink and also show the transistor implementation of the given load.

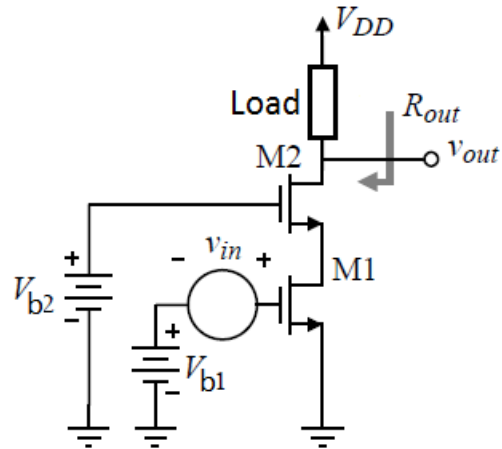


Fig. 1

[MM: 20]

- Q.2** (a) For the circuit shown in the **Fig.2(a)**, all the transistors have same threshold voltage and $L = 1 \mu m$, width $W_1 = 2 \mu m$ and $W_2 = W_3 = 40 \mu m$. $I_{Ref} = 25 \mu A$, calculate the current through each transistor. What is the lowest voltage at the output node (V_o) for which current source operation is possible. What are the values of g_m and r_o of Q_2 and Q_3 . Calculate the I_o / I_{REF} with and without considering channel length modulation. [MM: 10]
- (b) For the circuit shown in **Fig.2(b)**, identify the circuit of block **A**, block **B** and calculated the R_{out} and low frequency voltage gain (V_{out} / V_{in}). All the transistors have same threshold voltage and $L = 1 \mu m$, Width $W_0 = W_1 = 2 \mu m$ and $W_2 = W_3 = W_5 = 10 \mu m$, $I_{Ref} = 10 \mu A$. Take $g_{mb5} = 0.1 g_{m5}$. [MM: 10]

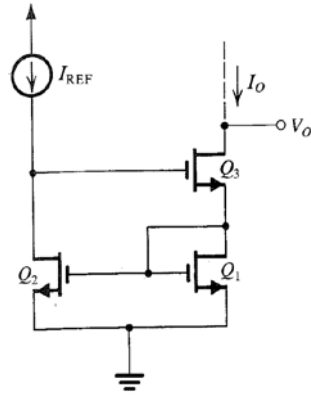


Fig. 2(a)

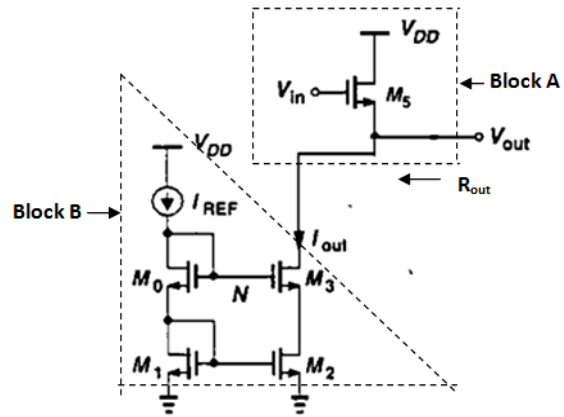


Fig. 2(b)

Q.3 Given Fig. 3, the overdrive voltage of all the transistors is equal $V_{ov}=0.3V$, $|I_{D1,2}| = 30\mu A$, .

Assume all transistors in saturation. ($\gamma = 0, \lambda \neq 0$)

- Calculate the low frequency voltage gain V_{out}/V_{in} When $R_1 = 0$ and $R_2 = \infty$.
- Calculate the output resistance R_{out} When $R_1 = 0$ and $R_2 = \infty$.
- Identify nature of sampling, mixing and hence the feedback topology ($R_{1,2} = 5k\Omega$).
- Calculate the loop gain ($A\beta$) and hence the closed loop voltage gain and the output resistance using concept of feedback. (Consider $R_{1,2} = 5k\Omega$)

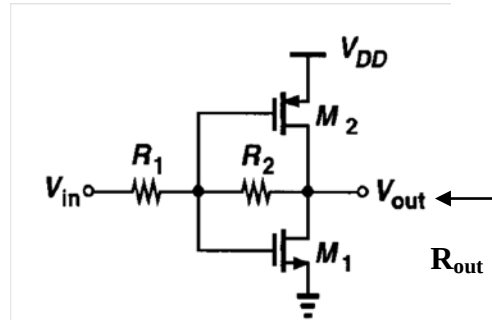


Fig. 3 [MM: 20]

Q.4 Fig.4 shows a two stage CMOS Op-Amp. Assume all the transistors are biased in saturation region.

- Find out the following:
 - dc value of current I_8 , I_5 and I_7
 - Low frequency differential voltage gain (v_O/v_{in}).
 - ICMR (Input common mode range)
 - Output resistance R_o .
 - Gain bandwidth in Hz.
 - Power dissipation
- Derive the expression for **RHP** (right half plane) zero which occurs in the scheme illustrated in dotted box region of the op amp. Also find out its value.
- Find out the expression for pole at the output node v_o . (Neglect parasitic capacitance).

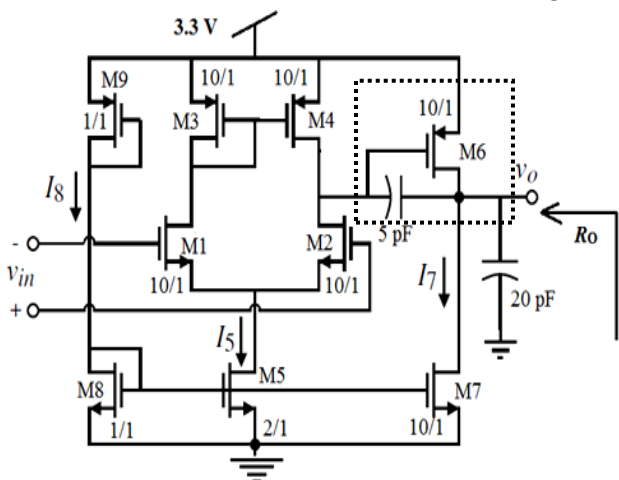


Fig. 4 [MM: 20]

*****End*****